Year 8

REVISION PACK

Unit 4

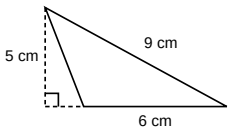
Name: _____

Class: _____

Unit 4 · Area of a triangle

Worked example: the area of a triangle

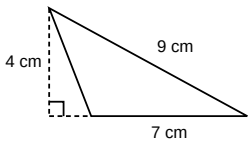
Find the area of this triangle.



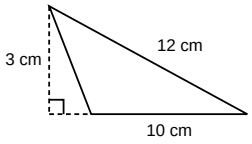
Step 1	Use base and height (at right angles)	base 6, height 5
Step 2	Half × base × height	$\frac{1}{2} \times 6 \times 5$
Step 3	Answer	$15\ \text{cm}^2$

Practice: the area of a triangle

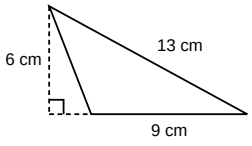
1.1 Find the area of this triangle.



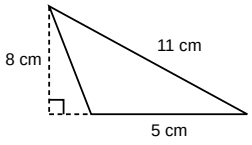
1.2 Find the area of this triangle.



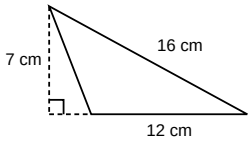
1.3 Find the area of this triangle.



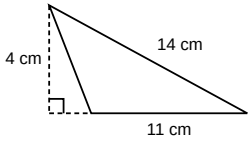
1.4 Find the area of this triangle.



1.5 Find the area of this triangle.



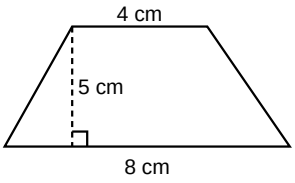
1.6 Find the area of this triangle.



Unit 4 · Area of a trapezium

Worked example: the area of a trapezium

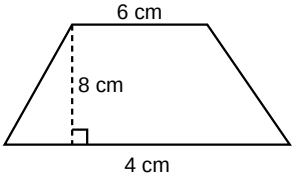
Find the area of the trapezium.



Step 1	Add the parallel sides	$4 + 8 = 12$
Step 2	Halve	$12 \div 2 = 6$
Step 3	Times the height	$6 \times 5 = 30$
Step 4	Answer	$30\ \text{cm}^2$

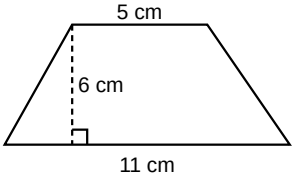
Practice: the area of a trapezium

2.1



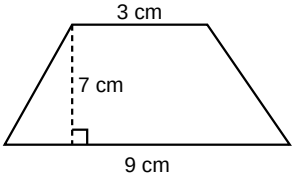
Find the area of the trapezium.

2.2



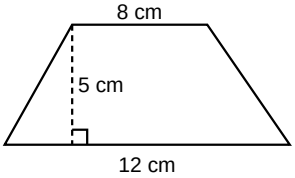
Find the area of the trapezium.

2.3



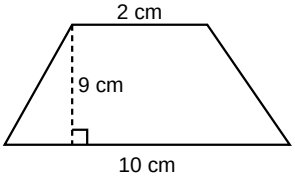
Find the area of the trapezium.

2.4



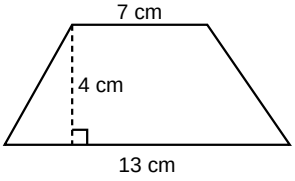
Find the area of the trapezium.

2.5



Find the area of the trapezium.

2.6



Find the area of the trapezium.

Unit 4 · Volume and surface area

Worked example: volume and surface area of a cuboid

A cuboid has edges of length 6 cm, 4 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

(a)	**Step 1**	Length × width × height	$6 \times 4 \times 2 = 48$
	Step 2	Answer	$48\ \text{cm}^3$
(b)	**Step 1**	Area of the 3 different faces	$6 \times 4 = 24,\ 6 \times 2 = 12,\ 4 \times 2 = 8$
	Step 2	Each face twice	$2 \times (24 + 12 + 8) = 88$
	Step 3	Answer	$88\ \text{cm}^2$

Practice: volume and surface area of a cuboid

3.1 A cuboid has edges of length 5 cm, 6 cm and 4 cm.

(a) Work out the volume. (b) Work out the total surface area.

3.2 A cuboid has edges of length 9 cm, 3 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

3.3 A cuboid has edges of length 7 cm, 7 cm and 5 cm.

(a) Work out the volume. (b) Work out the total surface area.

3.4 A cuboid has edges of length 10 cm, 6 cm and 3 cm.

(a) Work out the volume. (b) Work out the total surface area.

3.5 A cuboid has edges of length 8 cm, 5 cm and 4 cm.

(a) Work out the volume. (b) Work out the total surface area.

3.6 A cuboid has edges of length 11 cm, 4 cm and 2 cm.

(a) Work out the volume. (b) Work out the total surface area.

Worked example: litres and units of volume

A fish tank is a cuboid. The base measures 40 cm by 25 cm. Water is poured in to a depth of 10 cm.

(a) Work out the amount of water in litres. (b) Change 7 cm³ to mm³. (c) Show that 2.3 m³ = 2300 litres.

(a)	Step 1	Volume = l × w × h	$40 \times 25 \times 10 = 10\,000\ \text{cm}^3$
	Step 2	1 litre = 1000 cm³	$10\,000 \div 1000 = 10$
	Step 3	Answer	$10\ \text{litres}$
(b)	Step 1	1 cm³ = 1000 mm³	$7 \times 1000 = 7000\ \text{mm}^3$
(c)	Step 1	1 m³ = 1000 litres	$2.3 \times 1000 = 2300\ \text{litres}$

Practice: litres and units of volume

4.1 A fish tank is a cuboid. The base measures 45 cm by 30 cm. Water is poured in to a depth of 20 cm.

(a) Work out the amount of water in litres. (b) Change 50 cm³ to mm³. (c) Show that 4.6 m³ = 4 600 litres.

4.2 A fish tank is a cuboid. The base measures 60 cm by 30 cm. Water is poured in to a depth of 25 cm.

(a) Work out the amount of water in litres. (b) Change 8 cm³ to mm³. (c) Show that 3.2 m³ = 3 200 litres.

4.3 A fish tank is a cuboid. The base measures 50 cm by 40 cm. Water is poured in to a depth of 15 cm.

(a) Work out the amount of water in litres. (b) Change 12 cm³ to mm³. (c) Show that 1.75 m³ = 1 750 litres.

4.4 A fish tank is a cuboid. The base measures 80 cm by 25 cm. Water is poured in to a depth of 30 cm.

(a) Work out the amount of water in litres. (b) Change 4.5 cm³ to mm³. (c) Show that 6.8 m³ = 6 800 litres.

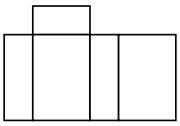
4.5 A fish tank is a cuboid. The base measures 35 cm by 20 cm. Water is poured in to a depth of 40 cm.

(a) Work out the amount of water in litres. (b) Change 30 cm³ to mm³. (c) Show that 0.9 m³ = 900 litres.

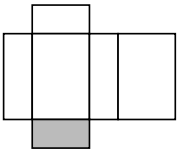
4.6 A fish tank is a cuboid. The base measures 70 cm by 30 cm. Water is poured in to a depth of 20 cm.

(a) Work out the amount of water in litres. (b) Change 2.6 cm³ to mm³. (c) Show that 4.25 m³ = 4 250 litres.

Worked example: completing a net and naming the solid

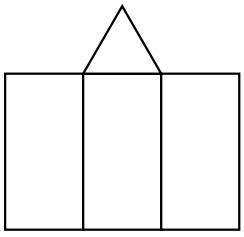


The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

(a)	**Step 1** Count the faces	5 rectangles, 1 missing	
	Step 2 Answer: add the shaded face		
(b)	**Step 1** 6 faces, all rectangles	cuboid	
(c)	**Step 1** Draw a box shape	show the hidden edges dashed	

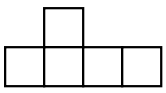
Practice: completing a net and naming the solid

5.1



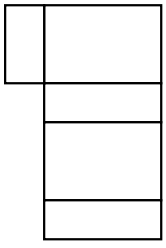
The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

5.2



The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

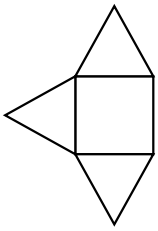
5.3



The diagram shows an incomplete net of a solid shape.
(a) Copy the net and add a face to complete it.
(b) What is the name of this solid?
(c) Sketch the solid.

Practice: completing a net and naming the solid (continued)

5.4



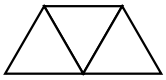
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

5.5



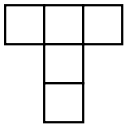
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

5.6



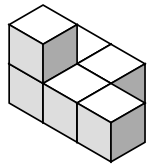
The diagram shows an incomplete net of a solid shape.

(a) Copy the net and add a face to complete it.

(b) What is the name of this solid?

(c) Sketch the solid.

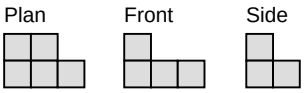
Worked example: drawing plans and elevations



front side

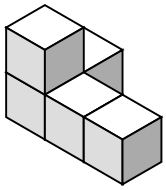
This model is made from 1 cm cubes. Draw the plan, the front elevation and the side elevation.

Step 1	Plan: look down from above	squares where there are cubes
Step 2	Front: look from the front	column heights $2, 1, 1$
Step 3	Side: look from the side	column heights $2, 1$
Step 4	Answer	



Practice: drawing plans and elevations

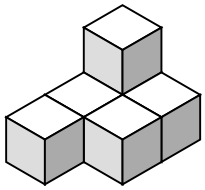
6.1



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

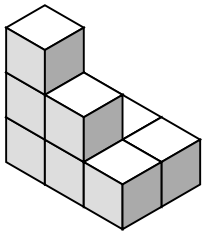
6.2



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

6.3

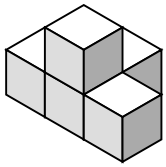


front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

Practice: drawing plans and elevations (continued)

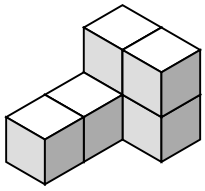
6.4



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

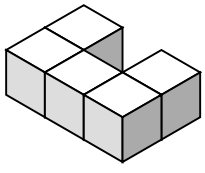
6.5



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

6.6



front side

This model is made from 1 cm cubes. On squared paper, draw the plan, the front elevation and the side elevation.

Worked example: a block with a hole

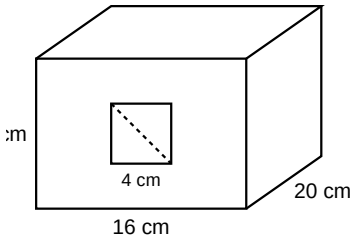
A block of wood is 10 cm wide, 8 cm high and 15 cm long. A square hole, with sides 2 cm, goes all the way through it.

(a) Work out the volume of the wood.
(b) Work out the total surface area of the wood.

(a)	Step 1	Whole block	$10 \times 8 \times 15 = 1200$
	Step 2	The hole	$2 \times 2 \times 15 = 60$
	Step 3	Block − hole	$1200 - 60 = 1140\ \text{cm}^3$
(b)	Step 1	Front and back	$2 \times (80 - 4) = 152$
	Step 2	Top and bottom	$2 \times 10 \times 15 = 300$
	Step 3	Two sides	$2 \times 8 \times 15 = 240$
	Step 4	Inside the hole	$4 \times 2 \times 15 = 120$
	Step 5	Add all the faces	$152 + 300 + 240 + 120 = 812\ \text{cm}^2$

Practice: a block with a hole

7.1

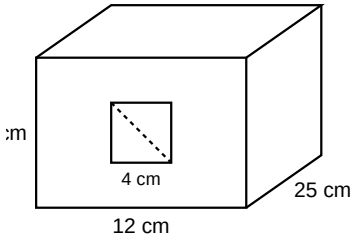


A block of wood has a square hole, with sides 4 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

7.2

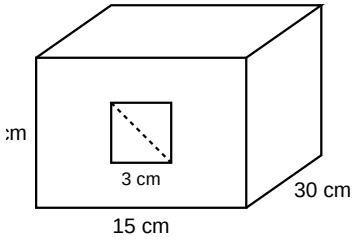


A block of wood has a square hole, with sides 4 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

7.3



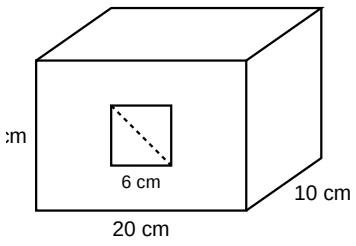
A block of wood has a square hole, with sides 3 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

Practice: a block with a hole (continued)

7.4

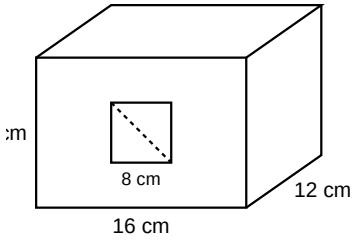


A block of wood has a square hole, with sides 6 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

7.5

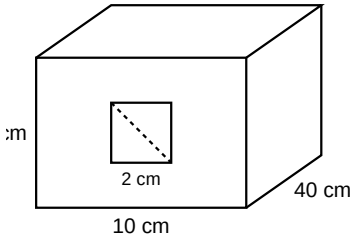


A block of wood has a square hole, with sides 8 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

7.6



A block of wood has a square hole, with sides 2 cm, drilled all the way through it.

(a) Work out the volume of the wood.

(b) Work out the total surface area of the wood.

Unit 4 · Area problems

Worked example: the area of a path

A square lawn has sides of 8 m. A path 50 cm wide goes all the way round it. Work out the area of the path in m².

Step 1	Change cm to m	$50\text{ cm} = 0.5\text{ m}$
Step 2	Side of lawn and path	$8 + 0.5 + 0.5 = 9$
Step 3	Big square	$9 \times 9 = 81$
Step 4	Lawn	$8 \times 8 = 64$
Step 5	Whole − lawn	$81 - 64 = 17$
Step 6	Answer	17 m^2

Practice: the area of a path

8.1 A square lawn with sides of 9 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

8.2 A square lawn with sides of 7 m has a path 100 cm wide all the way round it. Work out the area of the path in m².

8.3 A rectangular lawn 10 m by 6 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

8.4 A square lawn with sides of 15 m has a path 100 cm wide all the way round it. Work out the area of the path in m².

8.5 A rectangular lawn 5 m by 12 m has a path 50 cm wide all the way round it. Work out the area of the path in m².

8.6 A square lawn with sides of 3 m has a path 25 cm wide all the way round it. Work out the area of the path in m².

Worked example: a parallelogram and a triangle with the same area

A parallelogram has base 9 cm and height 4 cm. A triangle has the same area and a height of 6 cm. Show that the base of the triangle is 12 cm.

Step 1	Area of parallelogram: base × height	$9 \times 4 = 36$
Step 2	Triangle: half × base × height	$\frac{1}{2} \times b \times 6 = 36$
Step 3	½ × 6 = 3	$3b = 36$
Step 4	Divide by 3	$b = 12$
Step 5	Answer	$\text{the base is } 12 \text{ cm}$

Practice: a parallelogram and a triangle with the same area

9.1 A parallelogram has base 12 cm and height 5 cm. A triangle has the same area and a height of 8 cm. Show that the base of the triangle is 15 cm.

9.2 A parallelogram has base 15 cm and height 4 cm. A triangle has the same area and a height of 10 cm. Show that the base of the triangle is 12 cm.

9.3 A parallelogram has base 7 cm and height 6 cm. A triangle has the same area and a height of 12 cm. Show that the base of the triangle is 7 cm.

9.4 A parallelogram has base 10 cm and height 9 cm. A triangle has the same area and a height of 15 cm. Show that the base of the triangle is 12 cm.

9.5 A parallelogram has base 16 cm and height 3 cm. A triangle has the same area and a height of 6 cm. Show that the base of the triangle is 16 cm.

9.6 A parallelogram has base 9 cm and height 8 cm. A triangle has the same area and a height of 16 cm. Show that the base of the triangle is 9 cm.